



Yangon Technological University
Department of Electronic Engineering

PROGRESS SEMINAR

Design and Analysis of Real Time Multi-Robot Task Allocation System

Presented by:
Min Thant Kyaw (VI EC-1)

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Supervisor : Dr. Khin Kyu Kyu Win

Members : Daw Thi Thi Soe

Daw Zin Nwe Nyo Htun

Outline of Presentation

01. Introduction

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03. Current Progress

04. Tests & Results

05. Research Timeline

06. Future Works

07. Conclusion

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1. Introduction

- Modern warehouses rely on multiple autonomous robots.
- Tasks must be assigned efficiently to each robot.
- Centralized MILP finds globally optimal solutions.
- However, it becomes computationally expensive in real time.

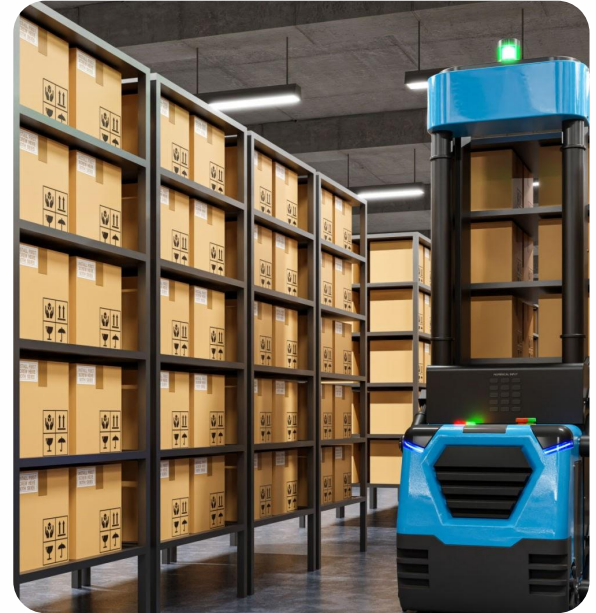


Figure 1.1: Autonomous Warehouse Robot

1. Introduction

1.1. Aim & Objectives

- To develop a learning-based multi-robot task allocation system to optimize the warehouse environment.
-
- To design a warehouse's layout and create a MILP-generated dataset.
 - To develop an XGBoost model for task allocation prediction.
 - To develop a local server for task management and processing.
 - To implement real-time communication between the server and robots.

2. Previous Work

2.1. Warehouse Layout Design

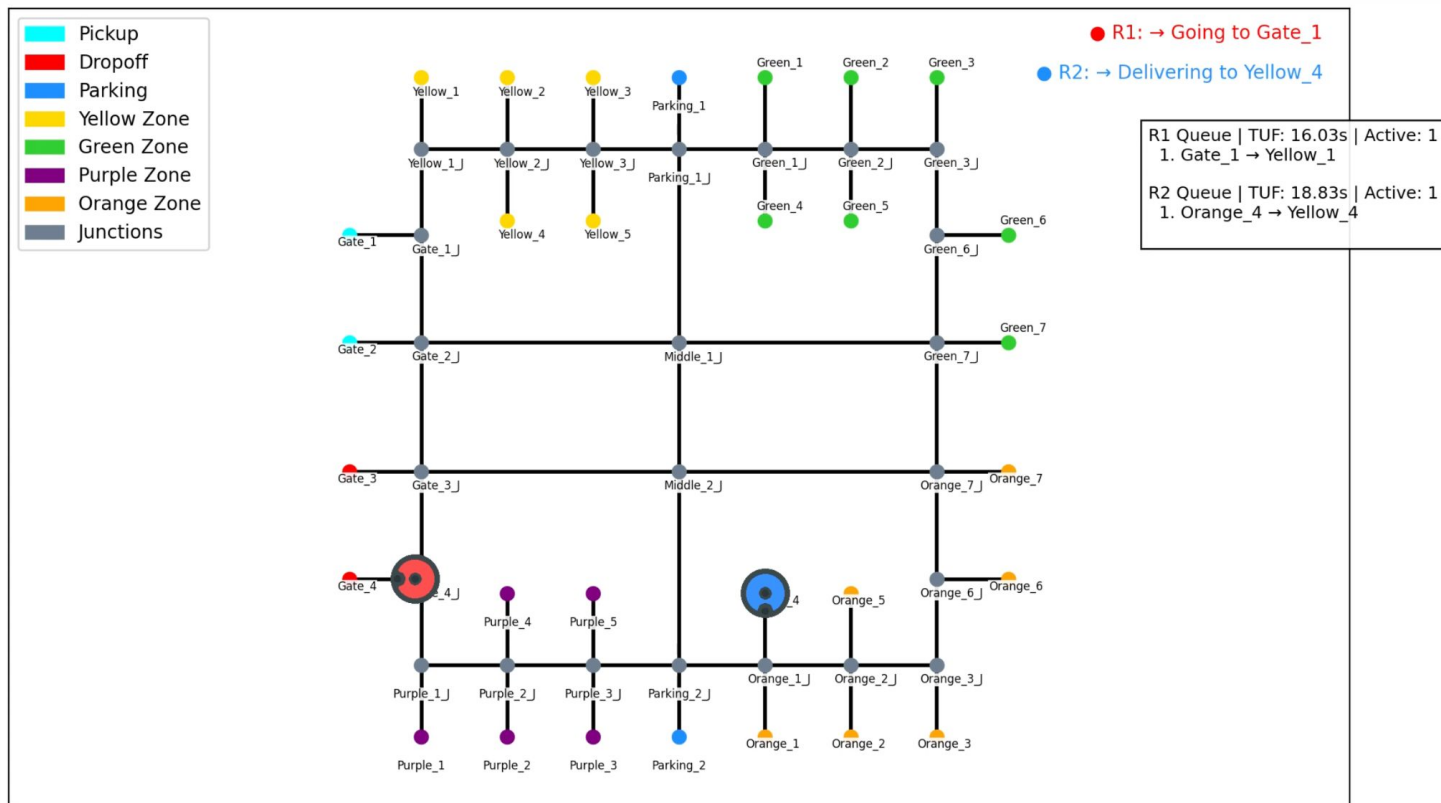
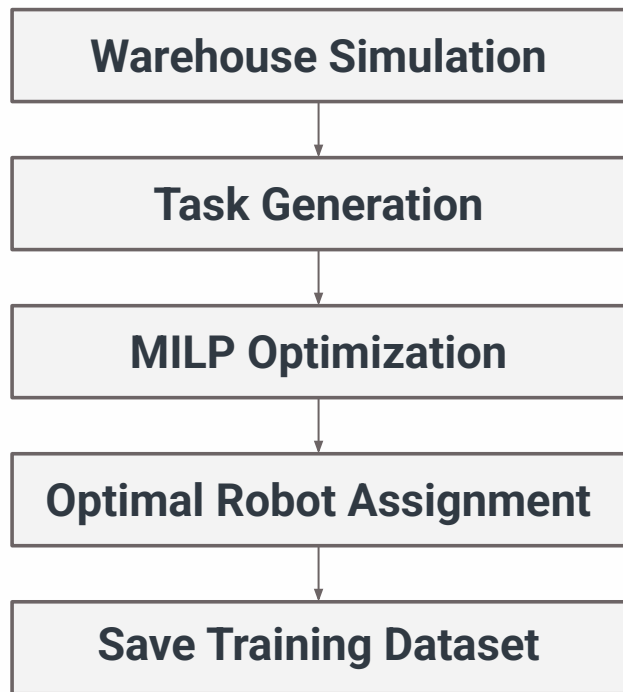


Figure 2.1: Warehouse Graph with NetworkX

2. Previous Work

2.2. MILP Dataset Generation



- Total dataset size: 20,019
- Number of features: 18
- Batch size range: 1-7
- Target: Assigned Robot
- 2 Output Classes: R0, R1

MILP = Mixed-Integer Linear Programming

Figure 2.2: MILP Dataset Generation Process

3. Current Progress

3.1. XGBoost Classifier Training

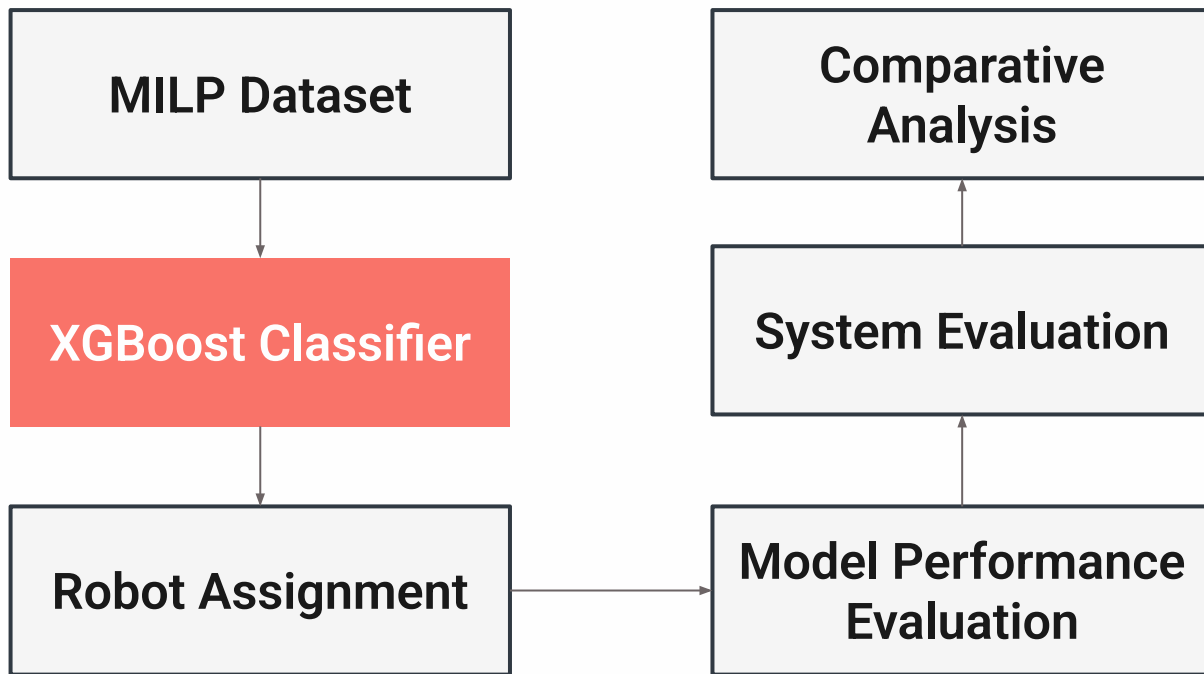


Figure 3.1: Workflow of XGBoost Classifier

3. Current Progress

3.1.1 Features Comparison

No.	Feature Category	This Research	Reference Papers
1	Pickup and Dropoff Locations	✓	✓
2	Physical Distance Between Nodes	✓	✓
3	Robot Coordinates	✓	✓
4	Robot Charging Percentage	✗	✓
5	Congestion Tracking	✗	✓
6	Temporal Constraints	✓	✓
7	Future Available Time	✓	✓
8	Relative Features	✓	✗

Table 3.1: Comparison of Features

3. Current Progress

3.1.2 Classifier Configuration

Batch Size:	1-7
Train/ Test Split:	75% / 25%
Objective:	Binary: Logistic
Evaluation Metric:	AUC-PR
Hyperparameter Optimization:	Hyperopt (100 Trials)

AUC-PR = Area Under Curve - Precision Recall

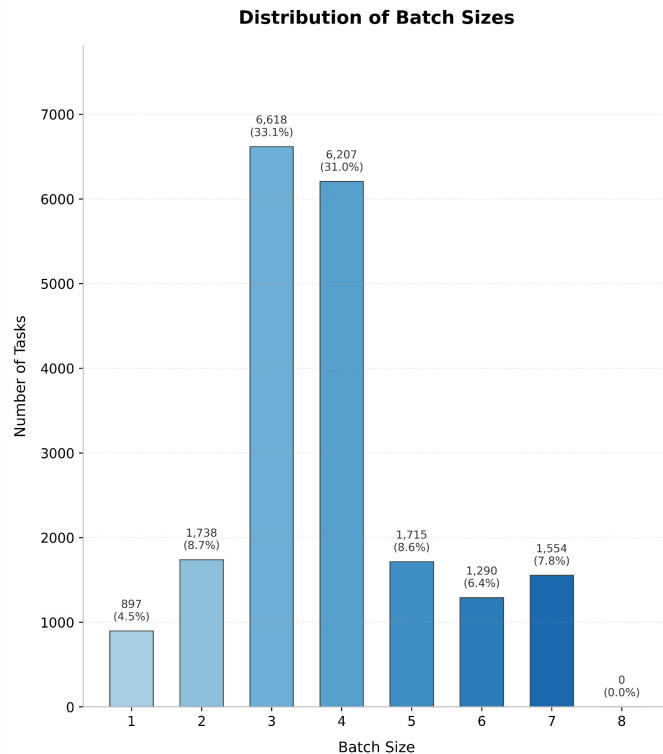


Figure 3.2: Batch Sizes Distribution

3. Current Progress

3.1.3 Classifier Performance (B1-7)

Precision
0.6592

Recall
0.7048

F1 Score
0.6812

Accuracy
0.6701

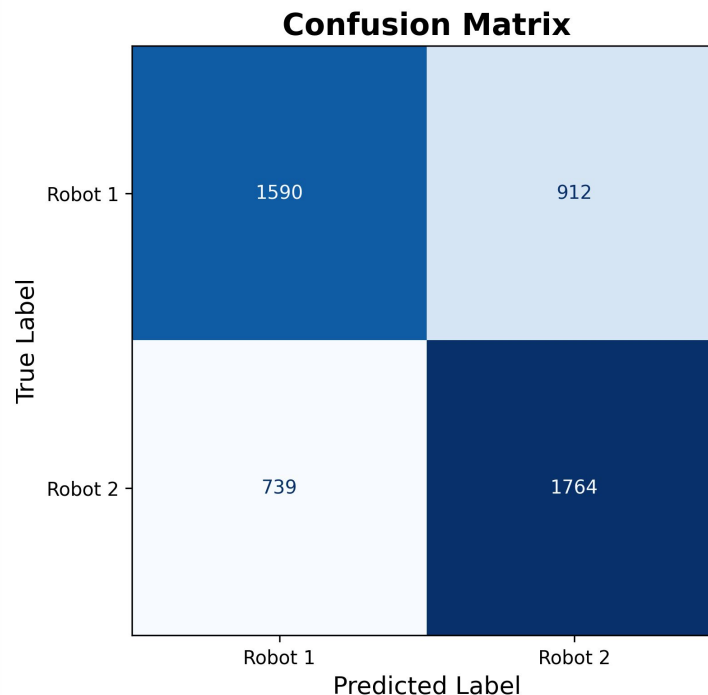


Figure 3.3: Confusion Matrix for XGBoost Classifier

3. Current Progress

3.1.4 Batch Size Investigation

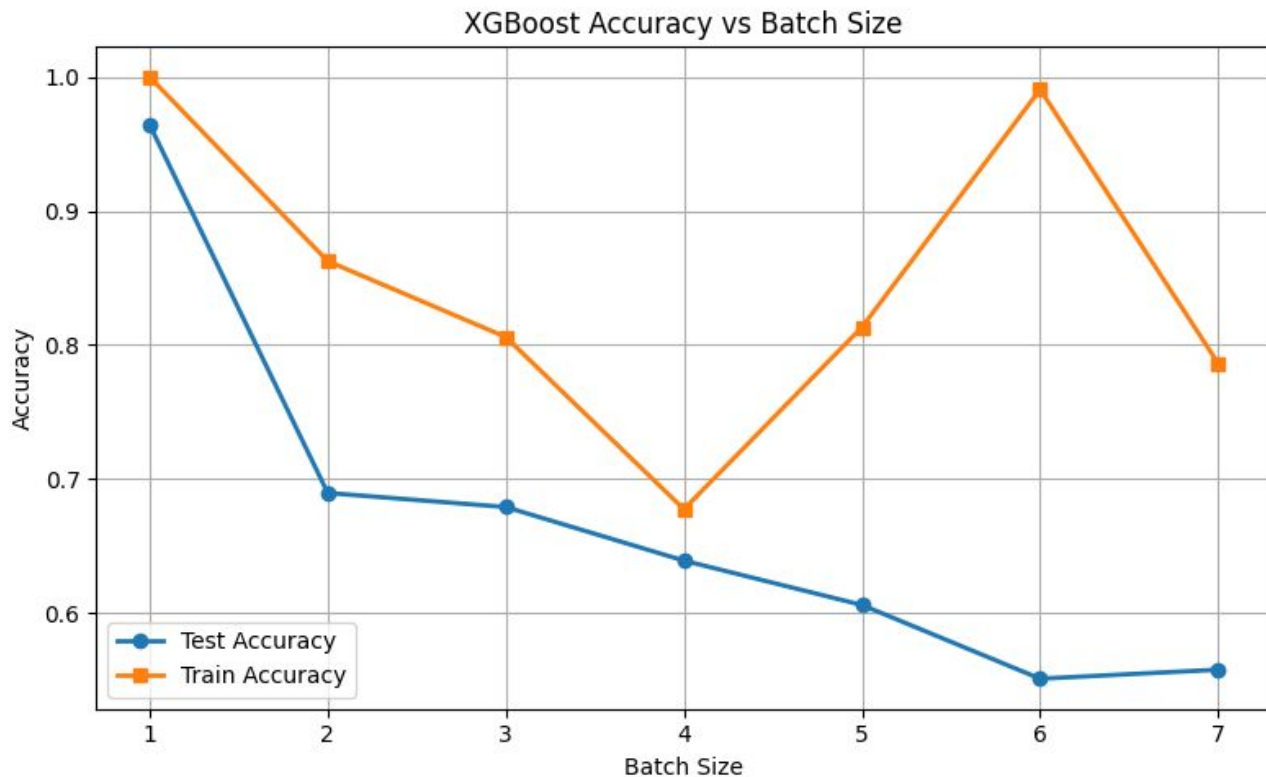


Figure 3.4: Investigation of XGBoost classification accuracy under different batch sizes.

3. Current Progress

3.2 Single-Task Dataset

Batch Size:	1
Train/ Test Split:	75% / 25%
Objective:	Binary: Logistic
Evaluation Metric:	AUC-PR
Hyperparameter Optimization:	Hyperopt (100 Trials)

AUC-PR = Area Under Curve - Precision Recall

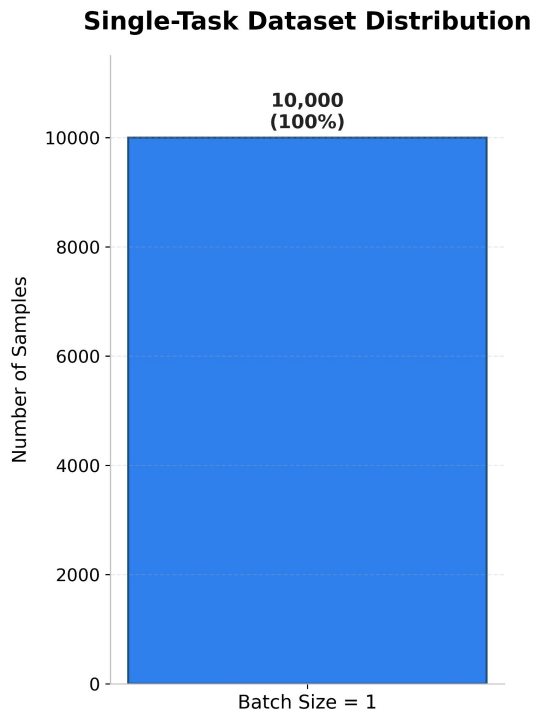


Figure 3.5: Single-Task Dataset Distribution

3. Current Progress

3.2.1 Classifier Performance (B1)

Precision
0.9904

Recall
0.9904

F1 Score
0.9904

Accuracy
0.9904

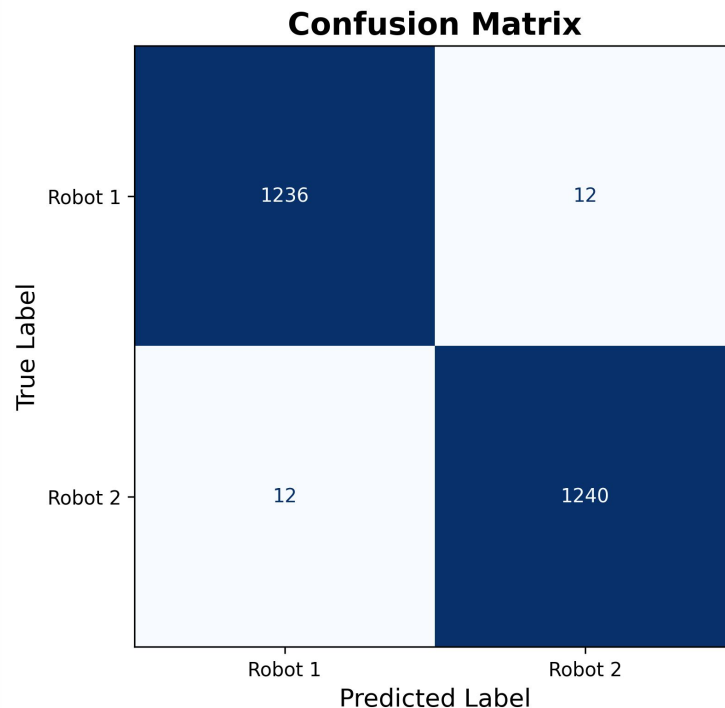


Figure 3.6: Confusion Matrix for XGBoost Classifier

3. Current Progress

3.3 Comparison of Models

Dataset	Precision	Recall	F1 Score	Accuracy
Mixed Batch Dataset (Batch Size: 1-7)	0.6592	0.7048	0.6812	0.6701
Mixed Batch Dataset (Batch Size: 1-7) With Hyperopt Optimization	0.6609	0.7048	0.6821	0.6715
Single Task Dataset (Batch Size: 1)	0.9904	0.9904	0.9904	0.9904
Single Task Dataset (Batch Size: 1) With Hyperopt Optimization	0.9936	0.9896	0.9916	0.9916

Table 3.2: Comparison of XGBoost models with different datasets and optimization

3. Current Progress

3.4 Feature Importance

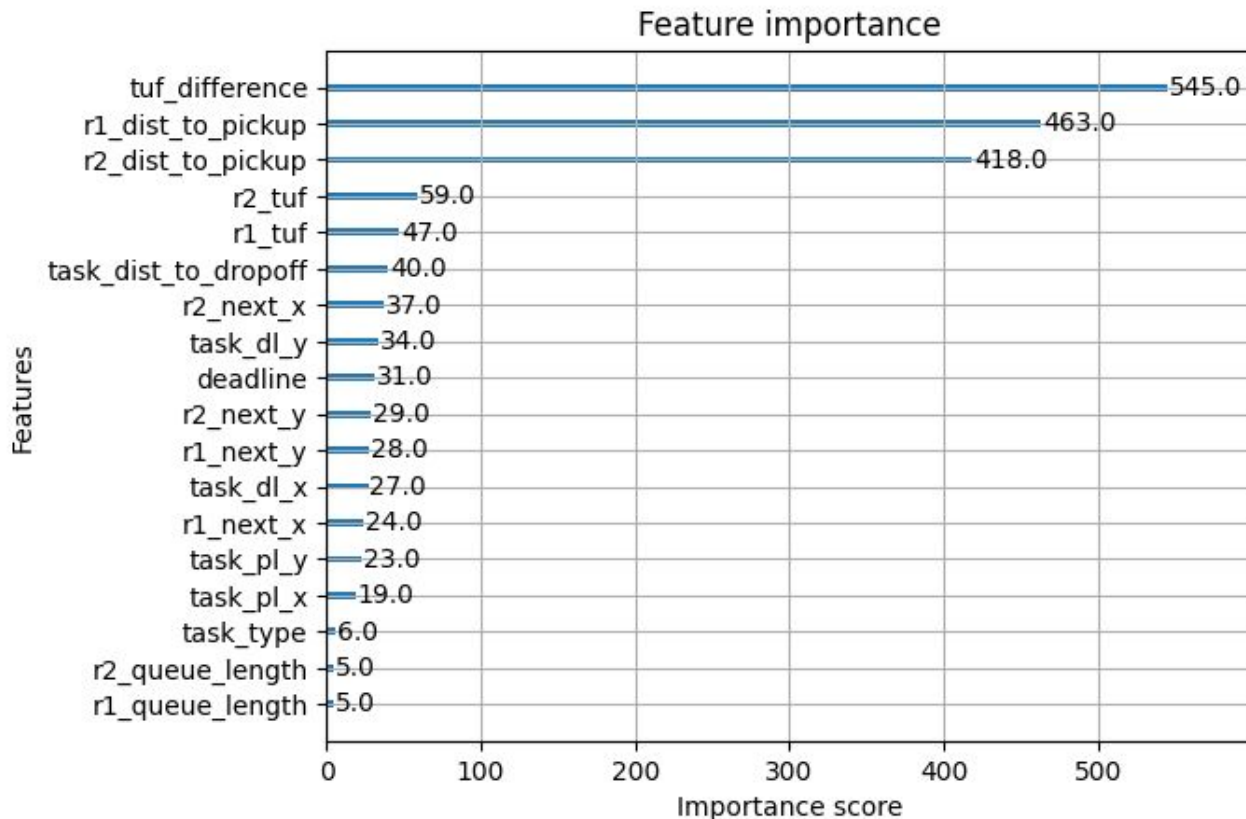


Figure 3.7: Feature Importance for XGBoost Classifier

3. Current Progress

3.5 System Evaluation

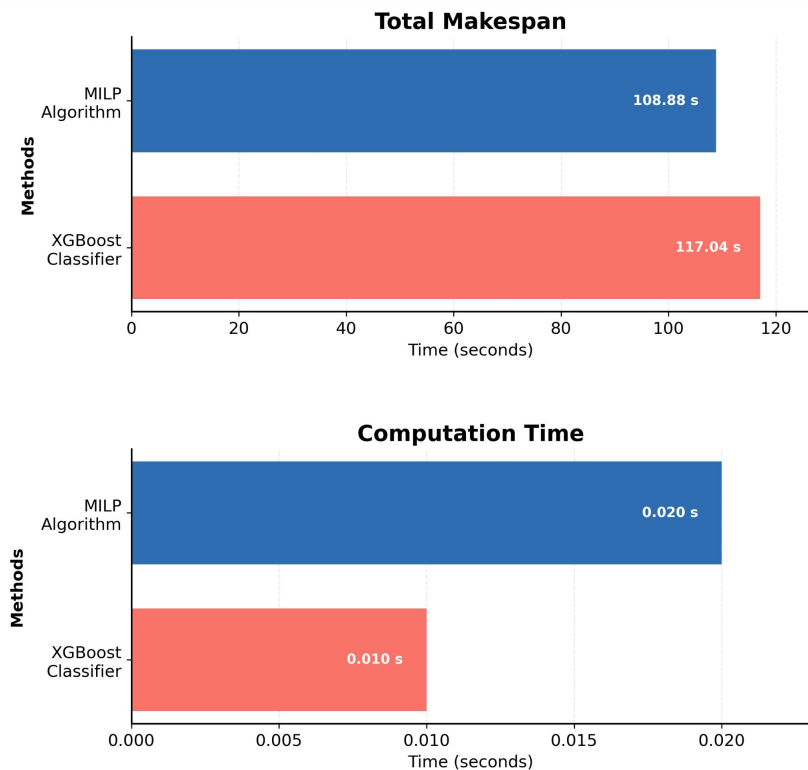


Figure 3.8: Comparison plots for fixed 10 tasks

Test Scenario

- Fixed **10** Tasks
- Same Pattern of Task Arrival

Makespan Gap:

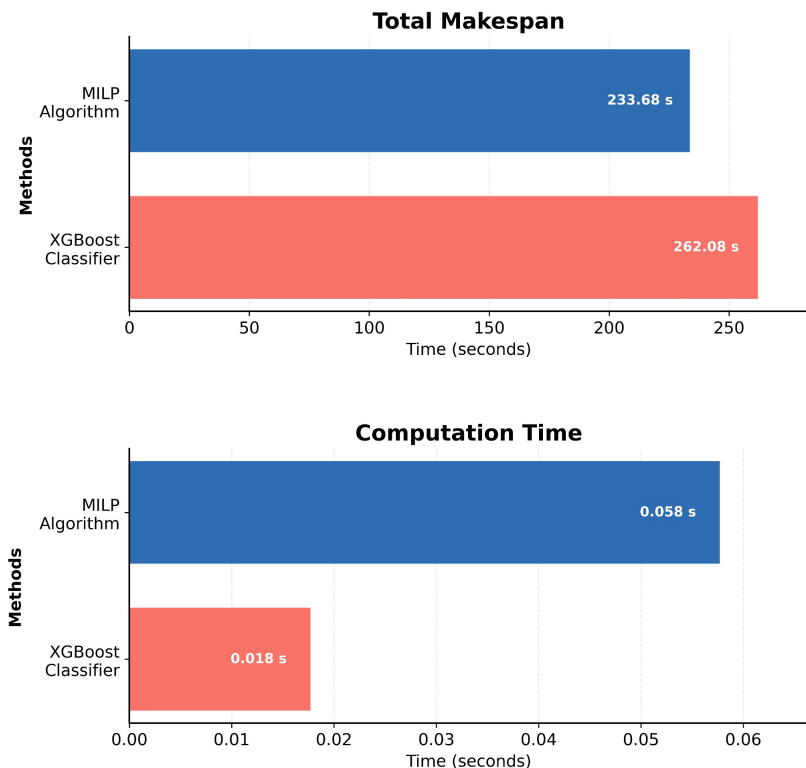
6.94%

Computation Time Reduction:

50%

3. Current Progress

3.5 System Evaluation (Cont'd)



Test Scenario

- Fixed **20** Tasks
- Same Pattern of Task Arrival

Makespan Gap:
10.83%

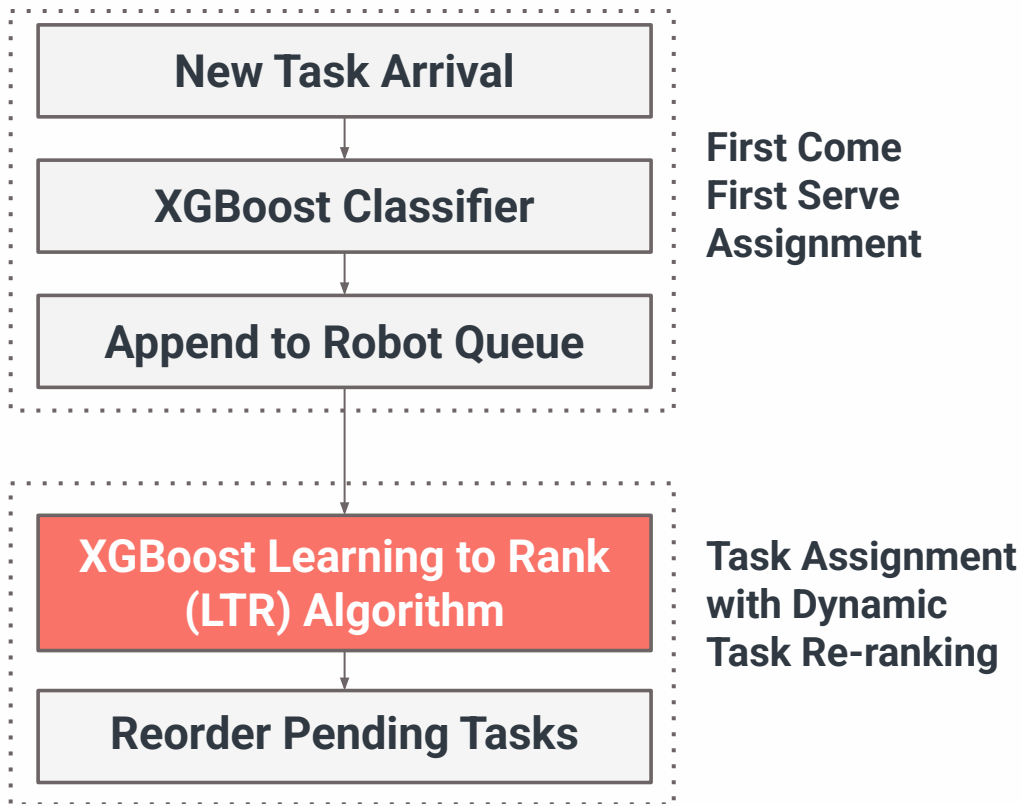
Computation Time Reduction:
68.96%

Figure 3.9: Comparison plots for fixed 20 tasks

3. Current Progress

- Makespan gap remains between XGBoost Classifier and MILP.
- MILP reduces makespan through task reordering.
- Developing XGBoost Learning-to-Rank for task re-ranking.

3.6 Modified Architecture



3. Current Progress

3.6.1 XGBoost LTR Dataset Features

Task Distance Relationship

Measures how close a task is to other tasks to estimate travel efficiency.

- Distance to Nearest Pickup
- Average Distance to Other Pickups
- Nearest Next Task Distance
- Average Next Task Distance

Deadline & Task Urgency

- Remaining deadline
- Deadline comparison
- Estimated lateness

Completion Prediction

- Estimated completion time
- Finish time if executed next
- Finish time if executed last

3. Current Progress

3.6.2 XGBoost LTR Dataset Analysis

Number of Samples: 10,7865

Number of Features: 80% / 20%

Objective: NDCG@1

Evaluation Metric: NDCG

NDCG = Normalized Discounted
Cumulative Gain

NDCG@1 = Normalized Discounted
Cumulative Gain at Top-1 Task

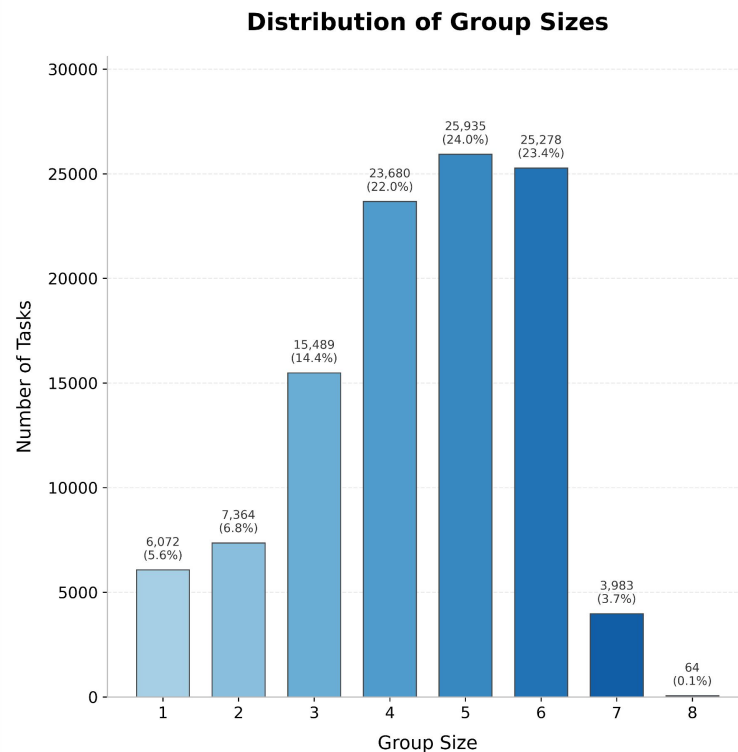


Figure 3.10: Group Sizes Distribution

3. Current Progress

3.6.3 XGBoost LTR Evaluation Metrics

$$NDCG@K = \frac{DCG@K}{IDCG@K} = \frac{\sum_{i=1}^{k \text{ (actual order)}} \frac{Gains}{\log_2(i+1)}}{\sum_{i=1}^{k \text{ (ideal order)}} \frac{Gains}{\log_2(i+1)}}$$

NDCG = Normalized Discounted Cumulative Gain

DCG = Discounted Cumulative Gain

IDCG = Ideal Discounted Cumulative Gain

Gains = The relevance (gain) score of the task

K = Number of top-ranked items

i = The position (rank index) of a task in the ranked list

3. Current Progress

3.6.4 XGBoost LTR Model Performance

NDCG:
0.8835

NDCG@1
0.7002

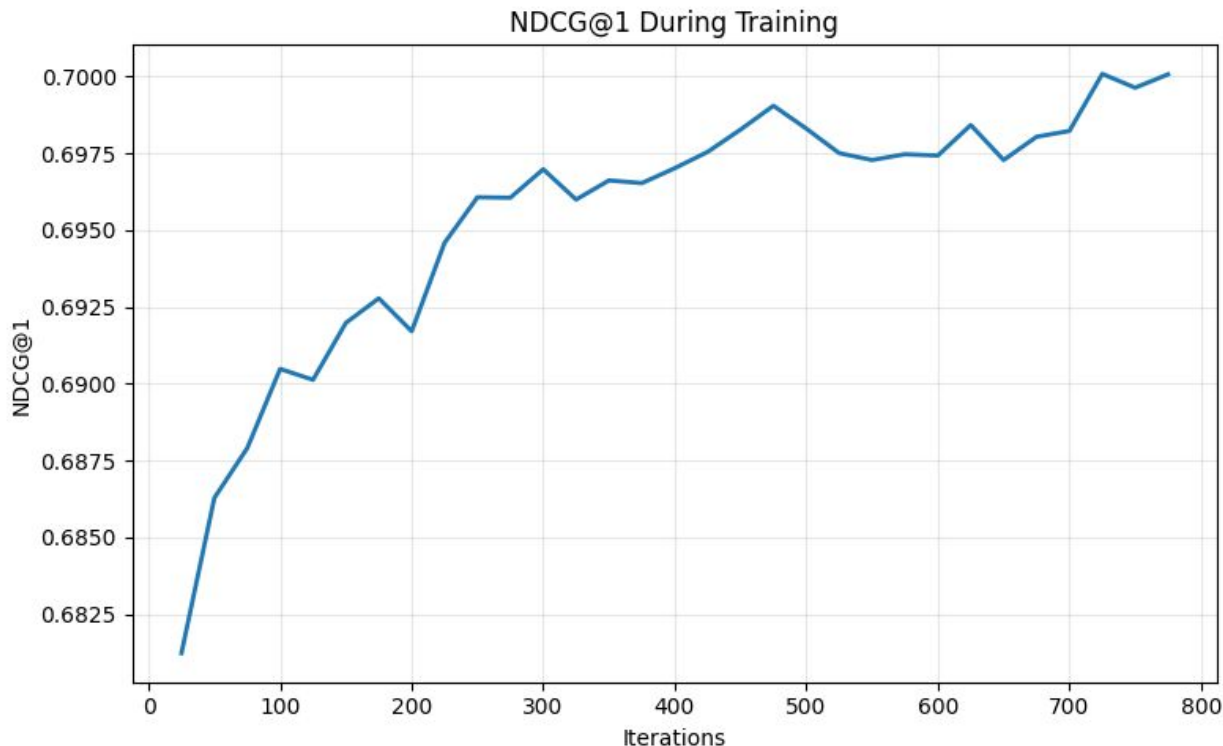
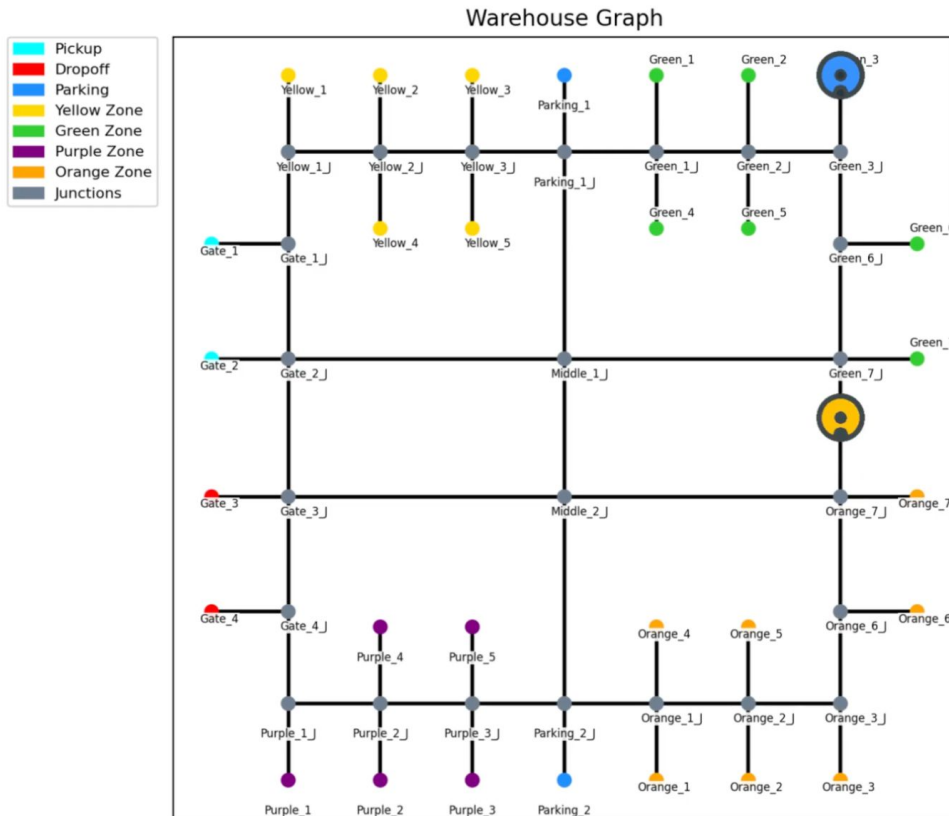


Figure 3.11: Plot for NDCG@1 During Training

4. Tests and Results

4.1 Simulation Video



R1 Queue

T13 LOCK
Orange_7 → Green_5

T11 ▼1
Gate_1 → Green_2

R2 Queue

T14 LOCK
Green_3 → Gate_3

T18 ▼1
Purple_3 → Yellow_4

4. Tests and Results

4.2 System Evaluation

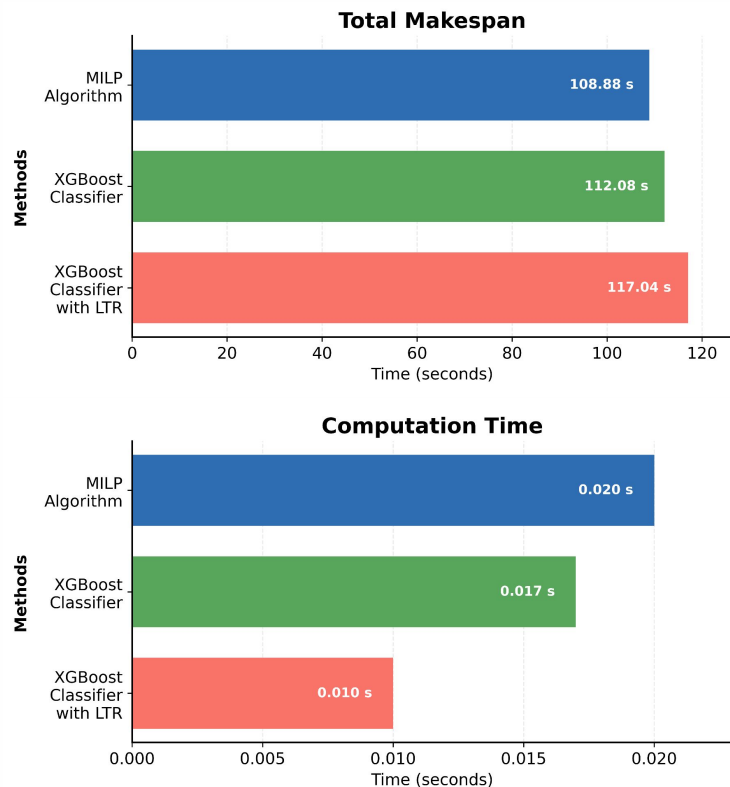


Figure 4.1: Comparison plots for fixed 10 tasks

Test Scenario

- Fixed 10 Tasks
- Same Pattern of Task Arrival

Makespan Gap:

Ranker
2.86%

First Come First Serve
6.94%

Computation Time Reduction:

Ranker
15%

First Come First Serve
50%

4. Tests and Results

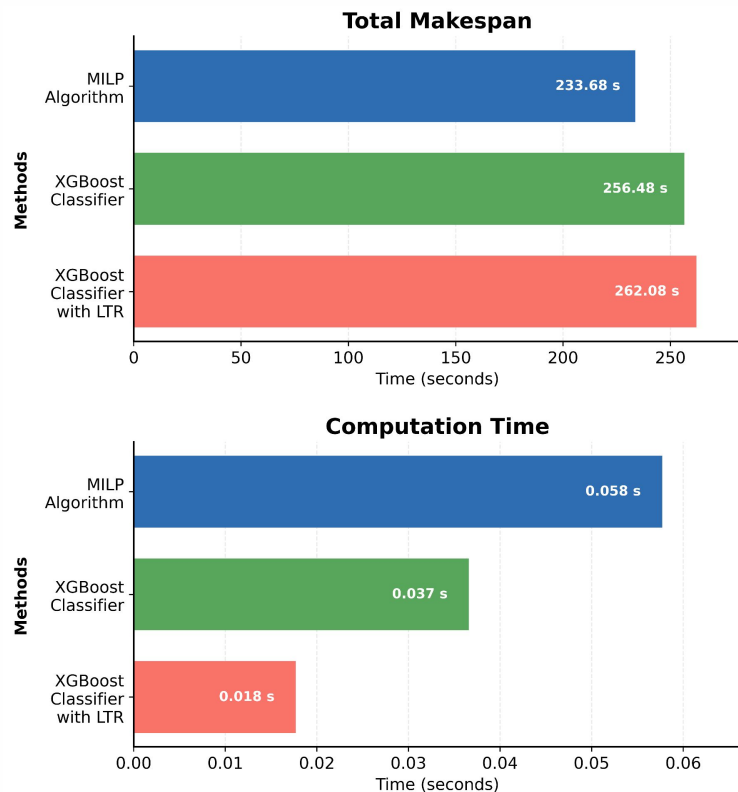


Figure 4.2: Comparison plots for fixed 20 tasks

4.2 System Evaluation (Cont'd)

Test Scenario

- Fixed **20** Tasks
- Same Pattern of Task Arrival

Makespan Gap:

Ranker	First Come First Serve
8.88%	10.83%

Computation Time Reduction:

Ranker	First Come First Serve
36.56%	68.96%

4. Tests and Results

4.3 Task Scalability Comparison

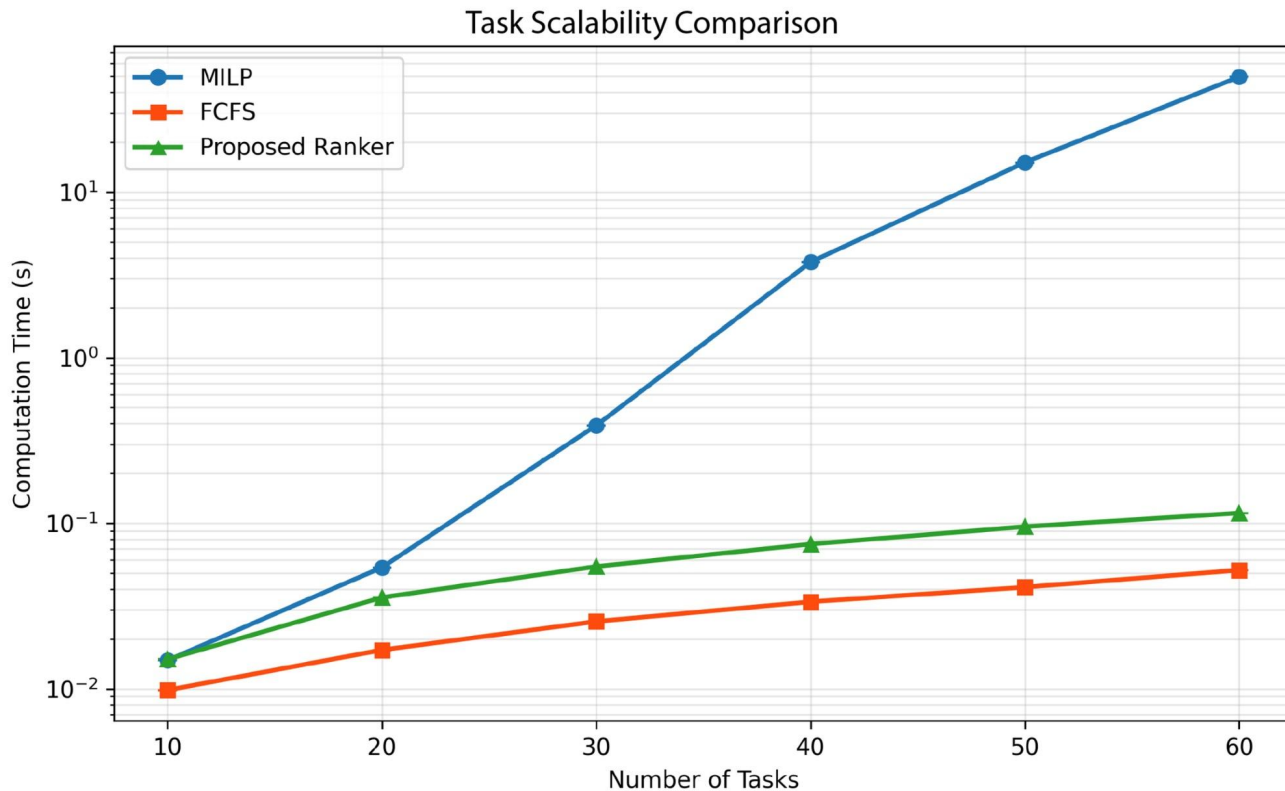
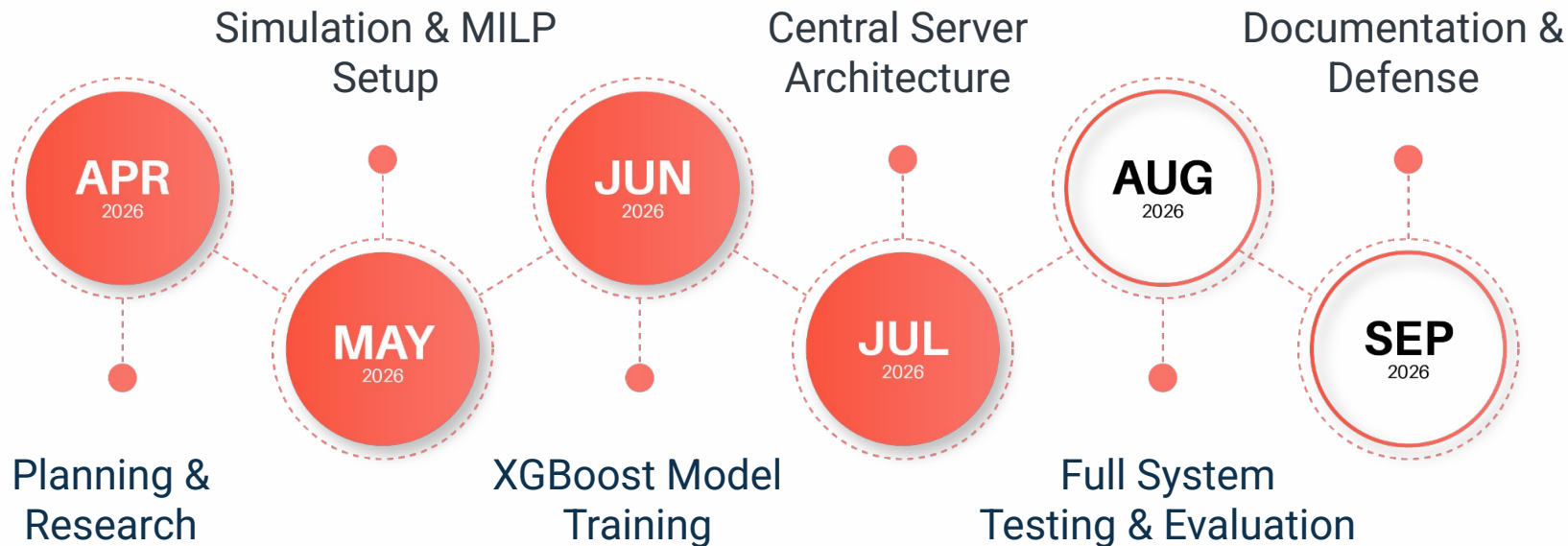


Figure 4.3: Task Scalability Comparison

5. Research Timeline



6. Future Work

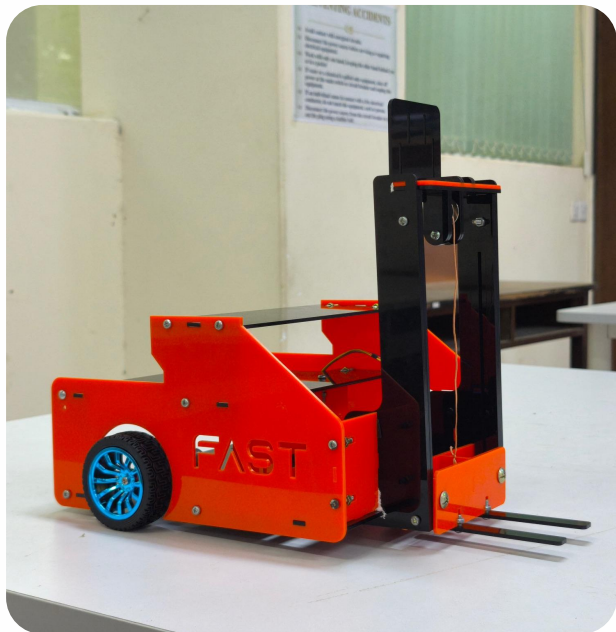


Figure 6.1: Warehouse Robot Design

- Integrate the classifier & Learning-to-Rank scheduler into the Warehouse Management System.
- Replace the simulation with real robots.
- Evaluate the framework on physical warehouse robots.

7. Conclusion

- XGBoost classifier successfully learned optimal robot assignment.
- Learning-to-Rank model developed for dynamic queue reordering.
- Overall scheduling performance significantly improved over First Come First Serve model.
- Future Integration into the warehouse management system and system evaluation with physical robots.

8. References

- [1] Pal A, Chauhan A, Baranwal M, Ojha A. , "Optimizing Multi-Robot Task Allocation in Dynamic Environments via Heuristic-Guided Reinforcement Learning", ECAI, 2024.
- [2] Park B, Kang C, Choi J. , "Cooperative Multi-Robot Task Allocation with Reinforcement Learning", Applied Sciences, 2022.
- [3] Putatunda S, Rama K. , "A Modified Bayesian Optimization based Hyper-Parameter Tuning Approach for Extreme Gradient Boosting", 2019.
- [4] Sharma A., "Listwise Learning to Rank with Deep Q-Networks", 2020.
- [5] Atay N., Bayazit B., "Mixed-Integer Linear Programming Solution to Multi-Robot Task Allocation Problem", Washington University in St. Louis, 2006
- [6] Athira K. A., Udayan J. D., Subramaniam U., "A Systematic Literature Review on Multi-Robot Task Allocation", ACM Computing Surveys, vol. 57, no. 3, 2024.

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